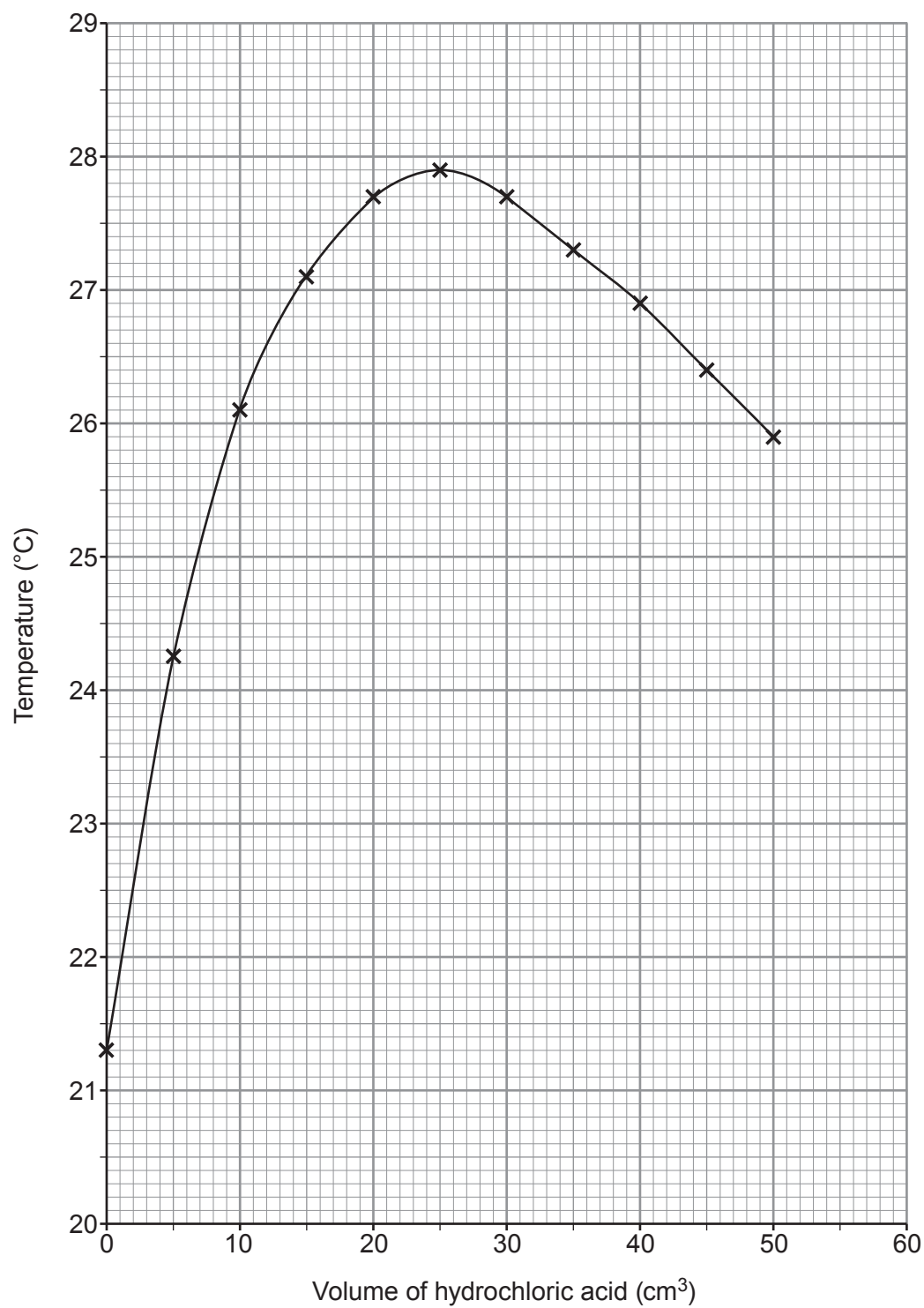


7. (a) The graph shows how the temperature changes when dilute hydrochloric acid is added gradually to sodium hydroxide solution.



The pH of the solution also changes as the hydrochloric acid is added to the sodium hydroxide.

Explain the temperature changes shown on the graph and relate these to the change in pH during the reaction. You do **not** need to include equations in your answer. [6 QER]

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- (b) The reaction between hydrochloric acid and sodium hydroxide produces sodium chloride and water.

Give the **ionic** equation to show how water is formed during this reaction. [1]

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END OF PAPER

7

